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1. Team Organisation & Structure

On 31st July, we officially met our team members for the first time. We used this opportunity to our advantage by exchanging not only contact numbers, but deciding a day for a meeting right off the bat. This meeting acted as a catalyst for great group dynamics. We were able to analyze the strengths and weaknesses of our counterparts to the best of our abilities. Having done so, the team was able to unanimously appoint a leader. Furthermore, we have two Quality Assurance Leads, one Technical Lead, one person assigned to take meeting minutes and one Testing Lead.

1.1. Structure

The team is structured in a way that ensures the work does not fall on only one person. This structure is a base our team decided to start with and update as required:

Anirudh Gupta	Group Leader
Yousef	Quality Assurance Lead
Khawaja	Technical Lead
Eric	Documentation (meeting minutes)
Edward	Testing lead
Amr	Quality Assurance Lead

Having such roles ensures that the team has someone to consult in case of confusion. A group leader ensures that deadlines are being met, work is being delegated fairly and resolves conflict. The group leader is also responsible for communication and guiding the team towards success. He keeps everyone accountable and reports any problems.

We also have two Quality Assurance Leads who ensure that every document/code the team delivers is up to the mark. If they feel something is not up to the mark, they can ask the team member responsible for a revised version. Having two leads also ensures that the leads are held accountable themselves.

We also decided to assign a person dedicated to taking meeting minutes. This ensures that the team has a meeting brief to refer to always, in case they missed something and want to revisit something.

A Testing Lead is a must have in a tech based consultancy. The role of the testing lead is to come up relevant test cases for a program. He can further assign people to test out the test cases thus made.

Technical Lead is one of the most important, if not the most important roles in this organization. He is responsible for giving the team a direction in terms of tech decisions.

1.2. Roles and Responsibilities

The 9th-bit has a full stack of developers, ranging from front end developers to back end developers. The project requires Web and App Developers, Software Tech Developers and AI Developers. The role and responsibilities are further discussed as follows:

ROLE	RESPONSIBILITIES	ASSIGNED TO
Web and App Developer	Frontend Developer, Django Template, UX/UI Design	Yousef
Software Tech Developer	Backend logic, Django setup, API integration, DB management, Deployment	Eric, Edward
ML/AI Developer	NLP Models, Data Processing, Data Scraping, Topic relevance scoring	Anirudh, Khawaja, Amr

2. Team Values & ACS Code of Professional Ethics

2.1. ACS Code of Ethics Overview

When we went through the ACS Code of Professional Ethics, it became clear that it is not just a checklist of do's and don'ts, but rather it is more like a guide for how to carry ourselves in our work. It puts a big emphasis on honesty, which for us means being upfront about what we can do, where we might need help and calling things out if we spot an issue. For example, if a task takes longer than expected or we run into a technical blocker, we would rather raise it early than keep it quiet and risk delays. It is also about being clear with clients, teammates and stakeholders, so there are no unnecessary conflicts.

Trustworthiness is another value we are committed to putting into practice. For us, it is not just about delivering on promise but being fully accountable for the work we produce. It is about taking ownership of our work, whether the outcome is a success or something that needs fixing. For example, if a feature fails testing, we own it, address it and learn from it. Additionally, it also means safeguarding the sensitive information we handle, whether that is user data, API keys or proprietary business logic from our sponsor. We aim to be consistent in our decisions, transparent about our capabilities and open about any skills we may need to build along the way.

Furthermore, respect is equally important and it applies both to people and the profession. Within the team, it means valuing every member's input, regardless of their level of expertise and fostering an environment where ideas can be shared without hesitation. For example, we make sure discussions stay open-ended, giving everyone the chance to contribute if they want to, without pressure [1].

Finally, respect for the profession means recognising our responsibility to create systems that are ethical, inclusive and beneficial to society. This includes designing solutions that consider accessibility, minimising bias in data handling and making environmentally conscious technical choices where possible. Beyond our project, it means contributing positively to the ICT community, whether that is through sharing knowledge, supporting peers or adopting best practices that uphold the reputation of ICT professionals.

2.2. Team Values

While the ACS Code of Ethics gives us a strong foundation, our team's values are shaped by the way we want to work together day-to-day. For us, openness comes first, if something is going off the track, we would rather address it early so we can fix it together instead of letting it become a potential concern. For example, if a backend integration is falling behind schedule, we prefer to bring it up in our next team check-in rather than wait until it becomes an issue. This approach prevents misunderstandings and supports a problem-solving culture where solutions are reached efficiently.

Commitment to follow-through is equally important. When someone commits to a deliverable, we all trust it will be done on time and to a high standard. Each team member is expected to deliver on their responsibilities reliably, while also communicating promptly if they require assistance. For instance, if a testing task reveals more bugs than expected, the testing lead will immediately notify the technical lead so priorities can be adjusted without delaying the timeline. This creates an environment where accountability is balanced with mutual support, which ensures no task is delayed due to avoidable bottlenecks.

Respect sits at the core of how we interact. We recognise and value contributions from all members, regardless of experience level and we strive to make decisions transparently, explaining the reasoning behind choices so everyone understands the direction [1].

Finally, we see this project as more than just a deliverable. It is a chance for all of us to walk away with more skills, confidence and perspective than when we started. Whether that is learning a new tool, improving our documentation game or just getting better at working in a team, growth is part of the plan.

3. Project Management Approach and Tools

3.1. Approach

The project runs in one-week iterations, following a simple "plan → build → demo" rhythm, with a short demonstration at the end of each week. Work proceeds in two lanes: Discovery, which covers API viability, basic ethics/ToS checks, and quick prototypes, and Delivery, which implements Django features, tests, and deployments. Tasks are kept small, typically a half day to two days, each with clear acceptance criteria and a named owner. Work moves across a straightforward board from To Do to Doing to Review to Done, and each person limits themselves to one or two in-progress tasks to reduce work in progress..

Lightweight but consistent, that's the quality assurance in my team. We have at least one reviewer for every pull request. Automated tests are run in the continuous integration pipeline. After that, there's a brief end-to-end check. This confirms that the major user flows still work, such as search returning results, and that searching for a profile still finds the profile.

We define what "done" means in a disciplined way that still feels natural to us. Because of that, we don't have very many issues. But when we do, the following conversation happens. The release path is predictable: a preview environment is created for each PR, if feasible; we merge to staging; and then we deploy to production. Production releases are hashed and tagged for easy rollback in the event that a release requires it. Basic data governance is in effect. We're reviewing ToS before we ingest any data; we're storing just the minimum necessary fields; and we're keeping a brief log of our major decisions and the key constraints that bound them.

3.2. Tools

Planning & Tracking

- **GitHub Projects:** For managing tasks, setting priorities, and tracking progress.
- **Labels:** Used for the purpose of work categorization (ai, backend, frontend).

Source Control & Code Review

- **GitHub:** The repository that houses the codebase; supports branching, pull requests, and reviews.
- **Branch Protections:** Guarantees that the reviews of push requests and passing CI checks are done before anything is merged.

Continuous Integration / Deployment (CI/CD)

- **GitHub Actions:** Used for automating the processes of testing and deploying code to both the development and staging environments.
- **Preview Deployments:** Would temporarily host versions of the app for testing changes in advance of the merge.

Runtime & Deployment

- **Fly.io:** Provides isolated environments for development and staging to host the Django application.

Database

- **MongoDB Atlas:** A cloud-based NoSQL database used for storing user queries and the processed results thereof.
- **Access Control:** Distinct staging and production databases; roles with minimum privileges.

Testing

- **pytest:** Automated unit as well as integration tests.
- **Playwright:** For end-to-end tests of the main user flows.
- **Postman:** For the API endpoint testing and documentation.

Design & Documentation

- **Google Drive:** For meeting notes, architecture documents, and decision logs.
- **Figma:** For wireframing and user interface (UI) design.

Communication Tools

- **Microsoft teams:** video conferences, presentations with the project supervisor and stakeholders, and virtual sprint reviews.
- **Discord:** Team collaboration, prompt questions, and share screens while developing.
- **Whatsapp:** To quickly notify people about important updates.
- **Gmail:** To formally speak to the sponsor to provide updates on progress, summarize meetings, and present important documents.

4. Communication Plan and Meeting Schedule

Effective communication is critical to ensuring the smooth progress of this project, maintaining stakeholder alignment, and preventing delays caused by misunderstandings. Our team will follow a structured communication framework that addresses both internal collaboration and external stakeholder engagement.

Internal Communication

The primary communication channel is **WhatsApp**, used for quick updates, task coordination, and urgent queries. WhatsApp will be used for urgent notifications or time-sensitive matters requiring immediate attention. **GitHub** will serve as the central hub for task updates, file sharing, and meeting scheduling, ensuring that project-related documents and discussions are organised and accessible to all members. All project-critical information, such as deadlines, action points, and decisions, is summarised and stored in a repository for easy reference. We also use **Discord** channels for deadlines and some brainstorming.

To maintain momentum, the team meets **every Friday at 3:00 PM** for a core weekly progress review via Zoom or in-person. In addition, **2–3 ad-hoc meetings** are held during the week as needed, particularly for resolving urgent matters or aligning on new developments. These meetings will cover progress updates, task allocation, and issue resolution. A meeting agenda will be shared at least 24 hours in advance, and **Eric** will document meeting minutes to provide a reference for all members, especially those unable to attend.

External Communication with Stakeholders

The Group Leader (**Anirudh Gupta**) will act as the primary point of contact for the sponsor (Robotics Marketers Pty Ltd) and the unit convenor. All formal project updates, documents, and milestone submissions will be communicated via email to ensure a professional and traceable record.

Project status reports will be delivered **bi-weekly** to the sponsor, containing summaries of completed tasks, upcoming deliverables, any risks or changes requiring stakeholder input, requirement clarifications, and feedback collection. External meetings are typically conducted via **Microsoft Teams**. The Technical Lead (**Khawaja**) may directly liaise with the sponsor's technical representatives if clarification on system architecture, API usage, or AI model integration is required, ensuring technical matters are resolved without unnecessary delays.

Meeting Schedule

Meeting Type	Participants	Frequency	Platform/ Location	Purpose
Weekly Progress Meeting	All team members	Every Friday – 3:00 PM	In-person / Zoom	Progress review, task allocation, issue resolution
Ad-hoc Team Meetings	Relevant team members	2–3 times per week	Zoom / WhatsApp call	Rapid alignment, urgent problem-solving

Sponsor Review Meeting	Group Leader, Technical Lead (as needed), Sponsor	Bi-weekly	Microsoft Teams / Email Summary	Milestone reporting, requirement clarifications
Unit Convenor Check-in	Group Leader, selected team reps	As scheduled	Email / iLearn	Academic compliance, guidance

Communication Principles

- **Transparency:** All progress and blockers are openly shared before and during meetings to enable early intervention; decisions are documented immediately after.
- **Responsiveness:** All messages acknowledged within 24 hours on working days.
- **Respect & Professionalism:** All interactions follow ACS ethical standards, ensuring constructive and respectful exchanges.
- **Traceability:** Critical updates and agreements are recorded in shared channels to ensure accountability and reduce miscommunication.

By following this communication plan, the team will ensure that all members and stakeholders are kept informed, aligned, and able to act promptly on project developments, fostering a smooth workflow from initiation to delivery.

5. Change Management and Conflict Resolution

Changes in project can occur in several form, including:

- **Scope changes:** this involves modification of project's objective
- **Schedule changes** (delays or accelerations)
- **Resource changes** (team member availability or budget adjustments)
- **Technical changes** (new technologies or tools introduced)

5.1. Managing Change

There are various forms that changes to a project may take and some of them include scope changes, schedule changes, resource changes, and technical changes each having its own strategies to deal effectively with them. With regard to **changes in scope**, one should consider using a controlled change process, through which new requirements should be recorded, the impact estimated, and stakeholders made aware of them before making them part of the project. For **Schedule changes**, The delay and speed-up of the schedule must be dealt with by re-evaluating the timeline, redistribution of the work, and benchmarking the milestones to ensure that the deadlines remain realistic. When there are **changes in resources**, including the availability of team members or budgetary limitations, it is possible to deal with them, by reevaluating priorities, shifting responsibilities, or searching other financial or human resources. **Technical changes** can be managed by bringing in new tools or technologies necessitate high-level evaluations of compatibility, training of team members and integration that is phased to avoid high levels of disturbance. In all types of changes, clarity of communication to the team and stakeholders as well as appropriate documentation will keep everything in line, minimise resistance and keep the project alive.

With such strategies, implemented in advance, project teams will handle surprise situations more easily, stay abreast of objectives, and minimize the possibility of costly operational delays. Adaptability can also be enhanced by creating regular review meetings, updating risk register and ensuring a culture open to change. By making them understand that change is the norm in project work, the team members will become stronger, cooperative, and problem solving which will enable the project work to move without losing its way or quality.

5.2. Conflict Resolution

Conflicts will be identified through the regular team meetings, allowing members to express their opinions on the current direction of the team. This allows for potential conflict to be identified before it escalates. Additionally, conflict can be identified through various signs, such as noticeable frustration during meetings, or repeated disagreements with current plans. After the conflict has been identified, the team has agreed on a structured method of managing the conflict:

1. **Open discussion:** The conflicting opinions will be first discussed during the next team meeting, allowing the members to explain their views.
2. **Consider options:** The team will propose potential solutions to be undertaken.
3. **Voting:** If there is no unanimous agreement on an available option, a vote will be held.
4. **Tie-Breaker:** In the event of a tie in voting, the leader will make the final decision.

6. References

[1] Australian Computer Society, *ACS Code of Professional Ethics: Adopted into the ACS National Regulations (NR4) on 14 March 2023*. Sydney, NSW, Australia: Australian Computer Society, 2023. [Accessed:13/08/2025].